Related questions:

Short Answer type questions:

Q1. Why quantum computers are supposed to perform the large scale calculations faster as compared to the classical computers.

Q2. Explain why high miniaturization of transistors is not good for classical computers.

Q3. Define Bloch sphere. Graphically represent the single qubit bases $|0\rangle$ and $|1\rangle$ on the Bloch sphere. Also give the possible range of values of angles 'θ' and 'ϕ'.

Q4. Define qubit. List out the different types of qubit representations.

Long Answer type questions:

Q1. Briefly explain the limitations of classical computers.

OR

Q1. Explain briefly the need of quantum computing.

Q2. List the differences between classical and quantum computers on the basis of fundamental features and performance and problem solving.

List of theory questions

1. Explain the **Bloch sphere representation** of a qubit with appropriate diagram.
2. Discuss the key aspects of quantum gates.
3. Describe in brief the matrix representation and action of different types of single-qubit gates on |0> and |1> qubit states.
4. Describe in brief the matrix representation and truth table of different types of multi-qubit gates on |0> and |1> qubit states.
5. Write the matrix representation and action of the (i) Pauli-X gate (ii) Pauli-Y gate and (iii) Pauli Z-gate on |0> and |1> qubit states.

Numericals

1. Apply the Hadamard gate followed by Pauli X-gate to |1>. Find the final state.
2. Apply the Hadamard gate followed by Pauli X-gate to |0>. Find the final state.
3. Apply the Pauli-X gate followed by Pauli Y-gate to |0>. Find the final state.
4. Apply the Pauli-Y gate followed by Pauli Z-gate to |0>. Find the final state.
5. Apply the Pauli-X gate followed by T-gate to |0>. Find the final state.

Sample Questions:

1. For two quantum systems, A and B, what does it mean for the systems to be entangled?
2. Using the Schrödinger's cat thought experiment, explain how the photon and the cat become entangled.
3. What is meant by a separable (product) state? Using the condition ad=bc, explain how one can determine whether a two-qubit state is entangled.
4. Starting from the initial state $|00\rangle$, describe step-by-step how applying a Hadamard gate followed by a CNOT gate generates an entangled Bell state.